Category Theory for Mathematical Physics — Exercise Sheet 8

Hopf algebras and quantum groups

Variant: Hints and answers

Conventions as in Lecture 8: an algebra is (H, μ, ι), a coalgebra $(H, \Delta, \varepsilon_H)$, the antipode S_H; P is the flip; $q \in \mathbb{C}^\times$ is generic, $q = A^2$. On $V_1 = \mathbb{C}^2$, with $e_1 = \uparrow$, $e_2 = \downarrow$, $K = \mathrm{diag}(q, q^{-1})$, $E = e_1 e_2^T$, $F = e_2 e_1^T$, and $\Delta(E) = K \otimes E + E \otimes 1$, $\Delta(F) = 1 \otimes F + F \otimes K^{-1}$, $\Delta(K) = K \otimes K$, $S_H(E) = -K^{-1}E$, $S_H(F) = -FK$, $S_H(K) = K^{-1}$. In this sheet F is that generator, never a functor (a fibre functor in Exercise 5 is written Φ, its structure maps $J_{g,h}$); the module map $V_1 \to V_1^*$ of Exercise 2 is written T. Braid generators carry their index, σ_i. The basis of $V_1 \otimes V_1$ is $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$, and B_q is the matrix of Theorem 8.9. In Exercise 6, *Integrability*'s anisotropy $\cos\eta$ is written Δ_{XXZ}, to keep Δ for the coproduct.

Exercise 1 (Functions and the group algebra). Let G be a finite group.

(i) Show that $\mathbb{C}^G$, with pointwise product, $\Delta(\delta_g) = \sum_{hk=g} \delta_h \otimes \delta_k$, $\varepsilon_H(\delta_g) = \delta_{g,1}$ and $S_H(\delta_g) = \delta_{g^{-1}}$, is a Hopf algebra.

(ii) Show that a $\mathbb{C}^G$-module is the same as a G-graded vector space, that the tensor product of Theorem 8.2 gives $V_g \otimes W_h$ the degree gh, and that the dual of Proposition 8.5 gives $(V_g)^*$ the degree g^{-1}: $\mathrm{Rep}(\mathbb{C}^G)$ is Vec_G.

(iii) Draw the bialgebra law Equation (8.1) for $\mathbb{C}[G]$ on $g \otimes h$, and conclude that $\mathrm{Rep}(\mathbb{C}[G])$ is $\mathrm{Rep}(G)$ with its tensor product.

Hint. (i) Check each axiom on the basis δ_g; $1 = \sum_g \delta_g$. (ii) The δ_g act as orthogonal idempotents summing to 1.

Answer. (i) Both sides of coassociativity are $\sum_{abc=g} \delta_a \otimes \delta_b \otimes \delta_c$; $\mu(S_H \otimes \mathrm{id})\Delta(\delta_g) = \sum_{hk=g} \delta_{h^{-1}}\delta_k = \delta_{g,1}1$. (ii) $V_g = \delta_g V$; $\Delta(\delta_k)$ is non-zero on $V_g \otimes W_h$ only for $k = gh$; δ_k acts on V^* as $\delta^T_{k^{-1}}$. (iii) Both sides send $g \otimes h$ to $gh \otimes gh$.

Exercise 2 (The antipode of U_q and the dual of V_1). (i) Draw the antipode identities Equation (8.2), with the counit and the unit, and check $\mu(S_H \otimes \mathrm{id})\Delta(x) = \varepsilon_H(x)1 = \mu(\mathrm{id} \otimes S_H)\Delta(x)$ for $x = E, F, K$.

(ii) By Proposition 8.5, x acts on V_1^*, in the dual basis e^1, e^2, by the transpose of $S_H(x)$. Write these matrices, and find the weights of V_1^* (the eigenvalues of K).

(iii) Show that $T\colon V_1 \to V_1^*$, $T(e_1) = e^2$, $T(e_2) = -q\,e^1$, is a module map, and that $(\mathrm{id} \otimes T^{-1})(\mathrm{coev}_{V_1}(1))$ is a multiple of the q-singlet $v = e_1 \otimes e_2 - q\,e_2 \otimes e_1$.

Hint. (i) It suffices to check generators, since S_H reverses products; $\varepsilon_H(E) = \varepsilon_H(F) = 0$, $\varepsilon_H(K) = 1$. (iii) Check $TX = X^*T$ for $X = E, F, K$.

Answer. (i) For instance $\mu(S_H \otimes \mathrm{id})\Delta(E) = K^{-1}E - K^{-1}E = 0$ and $\mu(S_H \otimes \mathrm{id})\Delta(F) = F - FKK^{-1} = 0$. (ii) $E^* = -q^{-1}e_2 e_1^T$, $F^* = -q\,e_1 e_2^T$, $K^* = \mathrm{diag}(q^{-1}, q)$: weights q^{-1}, q. (iii) $(\mathrm{id} \otimes T^{-1})(e_1 \otimes e^1 + e_2 \otimes e^2) = -q^{-1}v$.

Exercise 3 (R is not the braiding). On $\mathbb{C}[\mathbb{Z}_2] = \mathrm{span}(1, g)$, with $\Delta(g) = g \otimes g$, let $R = \frac{1}{2}(1 \otimes 1 + 1 \otimes g + g \otimes 1 - g \otimes g)$.

(i) Show that $R^2 = 1 \otimes 1$, $(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23}$ and $(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12}$, and explain why $R\Delta(x)R^{-1} = \Delta^{\mathrm{op}}(x)$ holds.

(ii) On the modules V_a (g acts by $(-1)^a$), show that R acts on $V_a \otimes V_b$ by $(-1)^{ab}$. Draw $c = P \circ R$ as in Equation (8.3), and say which of R and c is a morphism $V \otimes W \to W \otimes V$.

Hint. (i) Write $R = \frac{1}{2}\sum_{i,j}(-1)^{ij} g^i \otimes g^j$. (ii) Evaluate 1 and g on each factor.

Answer. (i) $R = \frac{1}{2}\sum_{i,j}(-1)^{ij} g^i \otimes g^j$ and the sums over i, j collapse; $\mathbb{C}[\mathbb{Z}_2]$ is commutative and cocommutative. (ii) $\frac{1}{2}(1 + (-1)^b + (-1)^a - (-1)^{a+b}) = (-1)^{ab}$; c is, R maps $V \otimes W$ to itself.

Exercise 4 (Hecke, Temperley–Lieb and Clebsch–Gordan). (i) Show that B_q has eigenvalue q on $\uparrow\uparrow$, $\downarrow\downarrow$ and $t = e_1 \otimes e_2 + q^{-1} e_2 \otimes e_1$, and $-q^{-1}$ on $v = e_1 \otimes e_2 - q\, e_2 \otimes e_1$; conclude the Hecke relation $(B_q - q)(B_q + q^{-1}) = 0$.

(ii) Show that $U = B_q - q = -q^{-1} v v^T$, that $U^2 = -(q + q^{-1})U$, and that $U_1 U_2 U_1 = U_1$ on $V_1^{\otimes 3}$ ($U_1 = U \otimes 1$, $U_2 = 1 \otimes U$).

(iii) Show that $\Delta(E)v = \Delta(F)v = 0$ and $\Delta(K)v = v$, and that $t = \Delta(F)(e_1 \otimes e_1)$. Normalise v and t, and take $q \to 1$: which Clebsch–Gordan states appear?

(iv) With $q = A^2$ and $R = \operatorname{diag}(A, A^{-1}, A^{-1}, A)\big(1 + (q - q^{-1})F \otimes E\big)$ (Theorem 8.9), compute $c = P \circ R$ and show $c = A^{-1} B_q$. What are the eigenvalues of c?

Hint. (ii) $v^T v = 1 + q^2$. For $U_1 U_2 U_1$, write $U_1 = -q^{-1}(v \otimes 1)(v^T \otimes 1)$ and U_2 likewise, and look at what is left between the outer factors. (iv) $F \otimes E$ has a single non-zero entry.

Answer. (i) Direct multiplication; the four vectors are a basis, so the relation holds on all of $V_1 \otimes V_1$. (ii) $U^2 = q^{-2}(1 + q^2) v v^T = -(q + q^{-1})U$. (iii) $v/\sqrt{1 + |q|^2} \to (\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2}$ (spin 0) and $t/\sqrt{1 + |q|^{-2}} \to (\uparrow\downarrow + \downarrow\uparrow)/\sqrt{2}$ (spin 1, $m = 0$). (iv) R has diagonal A, A^{-1}, A^{-1}, A and the entry $(\downarrow\uparrow, \uparrow\downarrow)$ equal to $A^{-1}(q - q^{-1})$; $PR = A^{-1} B_q$, eigenvalues A and $-A^{-3}$.

Exercise 5 (No fibre functor). A fibre functor Φ is a faithful $\mathbb{C}$-linear monoidal functor to $\mathrm{Vect}_{\mathrm{fd}}$ (Lecture 8). You may use that a $\mathbb{C}$-linear functor between semisimple categories preserves direct sums.

(i) Show that a fibre functor on Fibonacci would give an integer $n = \dim \Phi(\tau)$ with $n^2 = n + 1$, and conclude that there is none. Do the same for Ising.

(ii) The semion category has every FPdim equal to 1. Show with Sheet 3, Exercise 4, that it has no fibre functor either.

(iii) Why does this not contradict the existence of a fibre functor on Vec_G for $G = \mathbb{Z}_3$ with the pivotal structure $d = e^{2\pi i/3}$ of Lecture 4?

Hint. (i) $\Phi(\tau) \otimes \Phi(\tau) \cong \Phi(\tau \otimes \tau)$. (ii) Each $\Phi(\delta_g)$ is a line.

Answer. (i) $n^2 = \dim \Phi(1 \oplus \tau) = 1 + n$ has roots φ, $-1/\varphi$; Ising: $\dim \Phi(\psi)^2 = 1$, then $(\dim \Phi(\sigma))^2 = 2$. (ii) Identifying each $\Phi(\delta_g)$ with $\mathbb{C}$ makes Φ a functor of Sheet 3's kind, which needs $\omega = \delta j$. (iii) A fibre functor sees FPdim (here 1), not the pivotal d.

Exercise 6 ⋆. *The XXZ dictionary. Integrability* (its eq. (15.22)) writes the trigonometric R-matrix, in the basis $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$, as

$$R(\lambda) = \begin{pmatrix} \sinh(\lambda - i\eta) & & & \\ & \sinh\lambda & -i\sin\eta & \\ & -i\sin\eta & \sinh\lambda & \\ & & & \sinh(\lambda - i\eta) \end{pmatrix},$$

with $\Delta_{\mathrm{XXZ}} = \cos\eta$, η real and generic. Put $q = e^{i\eta}$.

(i) Let $D(\lambda) = \operatorname{diag}(e^{\lambda/2}, e^{-\lambda/2})$ and $\tilde{R}(\lambda) = (D(\lambda) \otimes 1)\, R(\lambda)\, (D(\lambda) \otimes 1)^{-1}$. Compute $\tilde{R}(\lambda)$. You may use that $\tilde{R}$ satisfies the same Yang–Baxter equation as R.

(ii) Show that $\lim_{\lambda\to-\infty}(-2e^{\lambda})\,P\tilde{R}(\lambda) = B_q$ and $\lim_{\lambda\to+\infty} 2e^{-\lambda}\,P\tilde{R}(\lambda) = B_q^{-1}$.

(iii) Show that without the gauge, $\lim_{\lambda\to+\infty} 2e^{i\eta}e^{-\lambda}PR(\lambda)$ satisfies the braid relation but not the Hecke relation.

(iv) Assemble the dictionary: q and Δ_{XXZ}; the over- and under-crossings with their scalars $A^{\mp 1}$; and the point $\eta \to 0$.

Hint. (i) $D \otimes 1$ multiplies $\uparrow x$ by $e^{\lambda/2}$ and $\downarrow x$ by $e^{-\lambda/2}$; only the two off-diagonal entries change. (ii) $\sinh x \approx -\frac{1}{2}e^{-x}$ as $x \to -\infty$; $2i\sin\eta = q - q^{-1}$.

Answer. (i) The entry $(\uparrow\downarrow, \downarrow\uparrow)$ becomes $-i\sin\eta\, e^{\lambda}$ and $(\downarrow\uparrow, \uparrow\downarrow)$ becomes $-i\sin\eta\, e^{-\lambda}$. (ii) At $-\infty$ the diagonal tends to $q, 1, 1, q$ and one off-diagonal entry to $q - q^{-1}$; at $+\infty$ to $q^{-1}, 1, 1, q^{-1}$ and $-(q - q^{-1})$; then P swaps the middle rows. (iii) The twisted swap with $1, 1$ on the corners and q off the diagonal; eigenvalues $1, 1, q, -q$, three distinct, so no multiple of it is quadratic. (iv) $q = e^{i\eta} = A^2$, $\Delta_{\mathrm{XXZ}} = (q + q^{-1})/2$; $\sigma_i \mapsto A^{-1}B_q$ $(\lambda \to -\infty)$, $\sigma_i^{-1} \mapsto AB_q^{-1}$ $(\lambda \to +\infty)$; $\eta \to 0$: $B_q \to P$.